

RSL-News

Large-Scale Dataset and Enhanced Lightweight Transformer Model for Gloss-Free Romanian SLT

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Introduction

- Sign languages
 - Primary means of communication for tens of millions
 - Fully formed natural languages
- No improvements in communication between Deaf and hearing people
 - Automatic SLT could remove this barrier
- ASLT - far behind other translation tasks
 - Gloss bottleneck: annotating 90 seconds takes experts 1 hour of work
 - Dataset capped at ~100 hours

Introduction

- Recent shift -> Gloss-free, E2E translation
- Can scale to thousands of hours from unused content from archives
- Still no datasets for Romanian
- **Our contributions:**
 1. **First** (Gloss-free) Romanian SLT dataset (among the largest)
 2. Improvements over a **lightweight (6.6M param.) SOTA** translation model
 - Bug Fixes
 - Décalage modelling
 - Lip-reading features
 3. First baselines for RSL translation & ablation studies

RSL-News Dataset

- A/V Law requires 30+ minutes of interpreted footage for news and debates!
- Derived from 3 TV stations:
 - Digi24, 11 AM News
 - ProTV, Morning News
 - PrimaTV, FOCUS

Table 3.1: Data source summary.

Channel	Date Range	Source	Episodes
Digi24	2019-11-12 – 2025-07-04	Website archive	1,200
ProTV	2021-01-20 – 2025-11-08	Website archive	1,117
PrimaTV	up to 2025-11-15	YouTube	578
Total			2,895

RSL-News Dataset

1. Scrape public web archives and YouTube playlists
2. Spatially crop "interpreter box"
3. Temporally segment active signing
 - For **Digi24**: label frames with pose estimation heuristic; train binary classifier; stitch the frame-level predictions using a state machine;
 - For **others**: binary search start / end interpreted block using facial recognition
4. Transcribe Romanian audio with *WhisperX*
5. Get sentence-level alignments with *WhisperX*
6. Normalize text & filter hallucinations



Figure 3.5: Representative frames showing all interpreters in the dataset.

RSL-News Dataset

Table 3.2: Per-channel statistics of the RSL-News dataset. “Singletons” are vocabulary items appearing exactly once. Lemma counts are computed using `spaCy` with `Stanza` [44] as the underlying NLP pipeline.

Channel	Ep.	Seg.	Sent.	Duration	Vocab	Lemmas	Singletons	% Single	Total Words	Avg Dur	Avg Len
PrimaTV	578	578	145,292	249h 45m	82,505	48,349	41,515	50.3%	2,370,825	6.19s	16.3w
Digi24	1,200	2,813	222,082	459h 02m	92,965	55,838	46,014	49.5%	4,172,034	7.44s	18.8w
ProTV	1,117	1,241	281,060	513h 51m	114,893	69,442	56,406	49.1%	4,612,740	6.58s	16.4w
Total	2,895	4,632	648,434	1222h 39m	180,892	111,974	93,547	51.7%	11,155,599	6.79s	17.2w

RSL-News Dataset

Table 3.3: Comparison of RSL-News against existing sign language translation datasets. Best BLEU-4 scores and statistics sourced from [5].

Dataset	Language	Vocab	Hours	Signers	Source	Best BLEU-4
PHOENIX-2014T [7]	DGS	3K	11	9	TV (weather)	26.75
CSL-Daily [34]	CSL	2K	23	10	Lab	21.61
How2Sign [4]	ASL	16K	79	11	Lab	15.5
iLSU-T [35]	LSU	38K	202	18	TV (news)	3.43
OpenASL [36]	ASL	33K	288	220	Web	13.21
YouTube-ASL [6]	ASL	60K	984	>2,500	Web	–
BOBSL [12]	BSL	77K	1,467	39	TV (general)	7.3
CSL-News [8]	CSL	5K	1,985	–	TV (news)	–
RSL-News (Ours)	RSL	180K	1,222	6	TV (news)	11.16

Methodology

- **SignFormer:** transformer-based, lightweight (6.6M params.), SOTA
- **X:** sentence video -> overlap. sliding windows -> I3D feat. extract. -> RGB feats.
- **Y:** text -> **SPM Unigram tokenizer** -> tokens

Methodology

SignFormer Bugs

1. *PositionwiseFeedForward* -> residual as $2x + \text{FFN}(\text{LN}(x))$ instead of $x + \text{FFN}(\text{LN}(x))$
-> gradient explosion
2. *ResidualConnectionModule* -> half-step residual as $0.5 \cdot \text{FFN}(x) + 0.5x$ instead of $0.5 \cdot \text{FFN}(x) + x$ -> vanishing gradients

=> No explosions but won't converge

Methodology

Décalage Modelling

- Interpreters translate with a natural delay => visual content is delayed w.r.t. the dataset timestamps => need explicit modelling

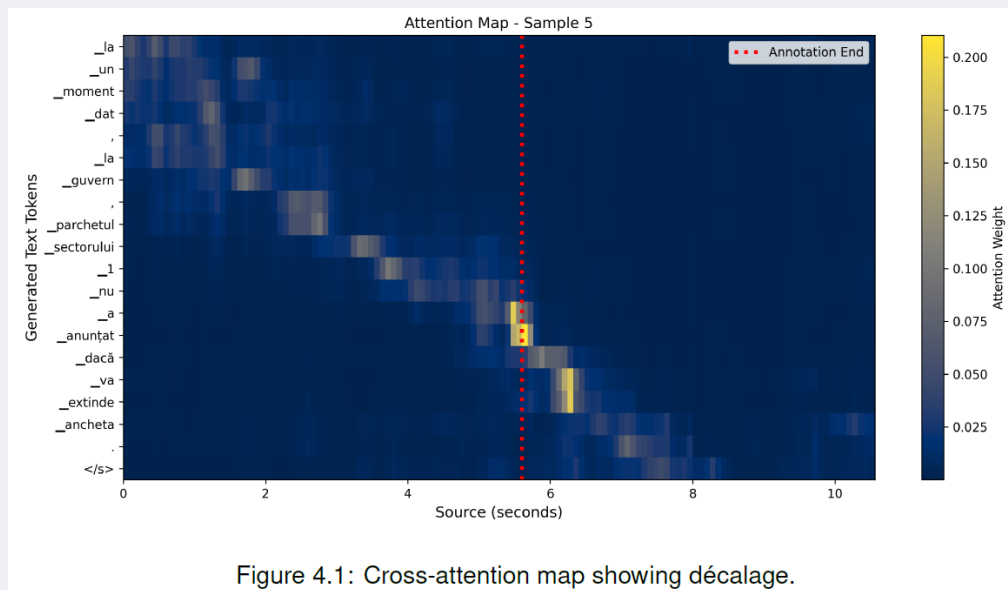


Figure 4.1: Cross-attention map showing décalage.

Methodology

Décalage Modelling

1. Extra content strategies:
 - Random (2, 4) seconds
 - Random (0, 5) seconds
 - Fixed 5 seconds
 - **Random (1, 5) seconds**
2. Confidence Ramps
 - Models uncertainty near boundaries
 - Scales input by confidence

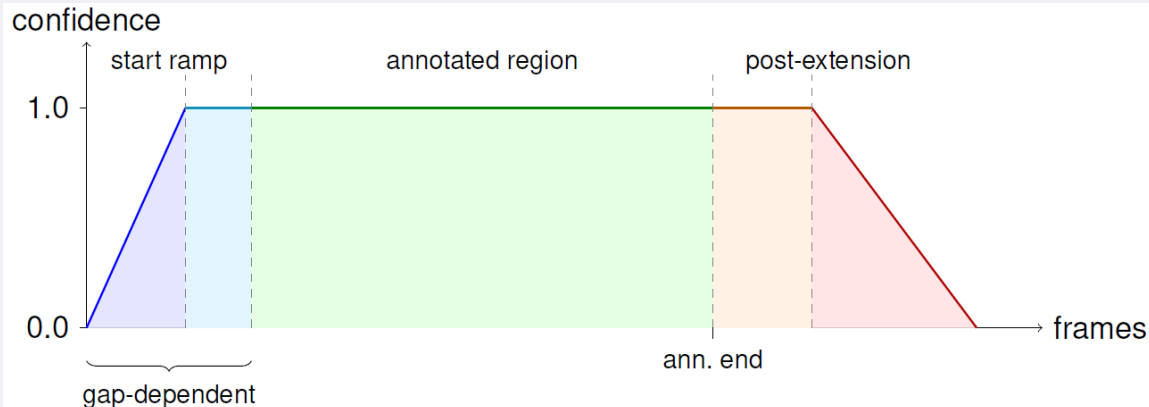


Figure 4.2: Confidence ramp structure.

Methodology

Mouthing Features

- Mouthings disambiguate between similar or hard to interpret signs
- Phonemized a 100+ hours Romanian Visual Speech Recognition dataset
- Fine-tuned *AutoAVSR*
 $\text{loss} = 0.9 \cdot \text{CTC} + 0.1 \cdot \text{CE}$ (to not overfit on Romanian)
- Used to inject features (logits) of signer's mouth per sample

Methodology

Mouthing Features

- Trained on Romanian, **not** RSL!
- **Spoken** sample: *Drumul de centură al orașului Călimănești din Vâlcea este în continuare închis.*
- Predicted phonemes from interpreter mouth:
numineStitun oraSalebeletfigZe zodels
- **Drum** → **num**: /d/ and /n/ -- same viseme; /r/ -- swallowed by /u/.
Visually, “drum” and “num” are indistinguishable.
- **Călimănești** → **neSti**: captured salient half of the word.
/S/, /t/, /i/ -- highly distinct lip configurations.
- **Oraș** → **oraS**: perfect phonetic match
- **Vâlcea** → **figZe**: /v/ and /f/ -- same viseme; /1/ (â) looks like /i/;
/tS/ (the ch in “Vâlcea”) and /gZ/ -- same puckering of lips

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Methodology

Mouthing Features

- Fused with sign feats. before encoder
- Full sign stream + phoneme gated (weighted) residual:

$$\alpha = \sigma(W_g \cdot \text{sgn} + b_g),$$
$$h = \text{sgn} + \alpha \cdot \text{LN}(\text{proj}(\text{phn}))$$

Results

Table 5.1: Progressive results across all configurations. “Rand (a, b)” means random post-extension between *a* and *b* seconds during training.

#	Configuration	BLEU-4	Epochs
1	Original SignFormer (both bugs present)	2.74	80
2	First bug fix only	2.86	82
3	First fix + rand (2, 4)s extra	3.57	97
4	Both fixes + rand (2, 4)s extra	6.56	103
5	Both fixes + rand (0, 5)s extra	9.19	209
6	Both fixes + flat 5s extra	9.31	205
7	Both fixes + rand (2, 4)s + ramps	8.95	61
8	Both fixes + rand (1, 5)s + ramps	9.98	188
9	Phonemes + flat 5s extra	10.01	199
10	Phonemes + flat 5s + 1 epoch ramps	10.30	200
11	Phonemes + rand (1, 5)s + ramps	10.76	200

Results

Reference

este vorba despre favipiravir, un antiviral care poate fi administrat în primele zile de boală pentru a preveni cazurile complicate și spitalizarea.

pentru că, repet, situația este una dramatică din punct de vedere demografic.

intervenția a fost dificilă, pentru că victima era prinsă în interiorul autoturismului.

Hypothesis

este vorba despre fentanilul, un medicament care poate fi prima zi de boală pentru a preveni situația complicată și internarea în spital.

pentru că situația este una dramatică din punctul meu de vedere.

intervenția a fost dificilă pentru că victima era încarcerată în caroseria distrusă.

Table 5.3: Samples showcasing entity hallucination, semantic shift, and creative paraphrasing.

Conclusions

Summary of Contributions

- **First** (Gloss-free) Romanian SLT dataset (among the largest)
- Improvements over a **lightweight (6.6M param.) SOTA** translation model
 - Bug Fixes
 - Décalage modelling
 - Lip-reading features
- Baseline results that prove the dataset as a SLT benchmark

Future Work

- JEPA-style training regime
- LLM as decoder
- Lip reader trained on pseudo glosses
- More diverse domain / conversational data
- Sign Language Generation

Thank You!

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